

18.06 Linear Algebra - Lectures 28, 29, 30
Textbook Problems - Sections 6.2, 7.1, 7.2, 8.1

◦ Section 6.2 Problem Sets

Lecture 22 already covered Section 6.2. ^{selected from} Q1-Q26
Selected questions 14, 24, 29, 38

~~14)~~

14) $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ not diagonalizable as ~~rank~~
 $\text{rank}(A - 3I) = \begin{bmatrix} 3-3(1) & 1 \\ 0 & 3-3(1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
 is 1 ✓

A is diagonalizable if it has two ~~distinct~~ eigenvectors.

{ ~~For~~ Change entries $\{a_{11}, a_{21}, a_{22}\}$ would provide an additional eigenvector and thereby make A diagonalizable. ✓

↳ change any except a_{12}

↳ bec A will have 2x different Eige

24) $A_{4 \times 4}$

$X \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$A = X \Lambda X^{-1}$

∴ $A_1 = X \Lambda_1 X^{-1}$

$A_2 = X \Lambda_2 X^{-1}$

$cA = c(X \Lambda X^{-1})$

$= X(c\Lambda)X^{-1}$

$A_1 + A_2 = X \Lambda_1 X^{-1} + X \Lambda_2 X^{-1}$

$= X(\Lambda_1 + \Lambda_2)X^{-1}$

Subspace as all have the same X ✓

When $X = I$ ∴ Subspace is all of $\mathbb{R}^4$ X diagonal matrices
 Dimension = 4 ✓ (matrices are 4×4)

29) $X \rightarrow$ diagonalizes $\{A, B\}$

$$A = X\Lambda_1 X^{-1}$$

$$B = X\Lambda_2 X^{-1}$$

Prove $AB = BA$

$$\begin{aligned} AB &= (X\Lambda_1 X^{-1})(X\Lambda_2 X^{-1}) \\ &= X\Lambda_1 (X^{-1}X)\Lambda_2 X^{-1} \\ &= X(\Lambda_1\Lambda_2)X^{-1} \end{aligned}$$

$$\begin{aligned} &\star \left\{ \begin{aligned} &\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \\ &\Downarrow \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} \\ &\Updownarrow \\ &\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} BA &= (X\Lambda_2 X^{-1})(X\Lambda_1 X^{-1}) \\ &= X\Lambda_2 (X^{-1}X)\Lambda_1 X^{-1} \\ &= X(\Lambda_2\Lambda_1)X^{-1} \end{aligned}$$

$$\Lambda_2\Lambda_1 = \Lambda_1\Lambda_2 \quad \star \checkmark$$

$$\therefore BA = X\Lambda_2\Lambda_1 X^{-1} = X\Lambda_1\Lambda_2 X^{-1}$$

$$\begin{aligned} \therefore AB &= X\Lambda_1\Lambda_2 X^{-1} = BA \\ AB &= BA. \end{aligned} \quad \checkmark$$

38) A similar to its Λ

$$A = B \Lambda B^{-1}$$

$\therefore B$ is the eigenvector ^(x) matrix and A must have n independent eigenvectors

Section 7.1 Problem Sets

1) $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{bmatrix} \begin{matrix} \times 1 \\ \times 2 \\ \times 3 \\ \times 4 \end{matrix} \left\{ \begin{matrix} \longrightarrow i \\ \downarrow j \end{matrix} \right.$

~~$r(A) = 1$~~ $r(A) = 1$ ✓

$B = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{matrix} +1 \\ +2 \\ +3 \\ +3 \end{matrix}$

$r(B) = 2$ ✓

$A = \sum_r u v^T$ $n \times m$ $n \times n$ $\boxed{}$

$A = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}^T$ $u^T = v^T = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$

~~$A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$~~ , $r = 1$
 $= \sigma_1 u_1 v_1^T$
 $\quad \quad \quad ? \times x$

~~u^T~~ $= \|u^T\|^2 = (1)^2 + (2)^2 + (3)^2 + (4)^2 = 1 + 4 + 9 + 16 = 30$
 $= \|v^T\|^2 = \text{same}$

~~u~~ $\|u\|^2 = \|v\|^2 = 30$
 $\|u\| = \|v\| = \sqrt{30}$

$\therefore \frac{u}{\sqrt{30}} = \frac{u}{\|u\|}; \quad \frac{v}{\sqrt{30}} = \frac{v}{\|v\|}$

$u_1 = \frac{u}{\|u\|} = \frac{u}{\sqrt{30}}; \quad v_1 = \frac{v}{\|v\|} = \frac{v}{\sqrt{30}}$

$\sigma_1 = \frac{1}{\|u\| \|v\|} = \frac{1}{\sqrt{30} \cdot \sqrt{30}} = 30$

divide A by $(\sqrt{30} \times \sqrt{30})$ so times by 30
 $A = \sigma_1 u_1 v_1^T = 30 \cdot \frac{u}{\sqrt{30}} \cdot \frac{v}{\sqrt{30}} = U \Sigma V^T$
 (1 col in U & V)

Q1) $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$

↳ Not orthogonal

↳ $\sigma\sigma$ does not give U 's and V 's of SVD

↳ Would need eigenvalues + vectors of $B^T B$ & $B B^T$

$B \rightarrow \text{rank } 2$

$B = U_1 V_1^T + U_2 V_2^T \rightarrow \text{Not the SVD}$

as pieces need to be orthogonal.

$\rightarrow$ Full $B^T B$ & $B B^T$

decomp required for SVD.

2) Walkthrough

All I singular values are $\sigma = 1$

↳ cannot leave out any terms $U_i \cdot V_i^T$ with making size I error

↳ None of the SVD pieces can be "left out" cannot compress

SVD of $I = (U)(I)(U^T)$
 $U = V$

$\sigma\sigma$ natural choice

$U \Sigma V^T = I I I$

↳ Could choose any U (orthogonal matrix)

↳ Eigenvectors of I are not unique - many choices.

Compressability
is about Singular Value Decay

$\rightarrow$ Compress when $\sigma_1 \gg \sigma_2 \gg \dots$
so tail is discarded

$\rightarrow I \rightarrow$ zero decay $\rightarrow$ flat spectrum

$\left\{ \begin{array}{l} I \rightarrow \text{rank } n \rightarrow \text{so need all } n \text{ pieces} \\ \rightarrow \text{also every value} = 1 \rightarrow \text{so no ordering by importance} \end{array} \right.$

$$6) A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \quad \text{A}$$

$$\begin{aligned} \det(A - \lambda I) &= (2-\lambda)(2-\lambda) - (1)(4) \\ &= 4 - 4\lambda + \lambda^2 - 4 \\ &= \lambda^2 - 4\lambda = \lambda(\lambda - 4) \\ \therefore \lambda &= 0 \text{ or } 4. \checkmark \end{aligned}$$

~~Singular Values: $\left\{ \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$~~

$$A^T A = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$$

eigenvalues: 0, 25

walkthrough
Eigenvectors

v_1, v_2 of $A^T A$:

$A^T A$ ~~are~~ is a symmetric matrix.
 $\therefore A^T A$ eigenvectors are orthogonal.

$$\begin{aligned} \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} &= 25 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} &= 0 \begin{bmatrix} 1 \\ -2 \end{bmatrix} \end{aligned} \quad \begin{bmatrix} 50 \\ 25 \\ 0 \\ 0 \end{bmatrix}$$

$\lambda_1 v_1$
 $\lambda_2 v_2$

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

$\therefore$ symmetric

$$\text{also } x^T (A^T A) x = \|A x\|^2 \geq 0$$

$\therefore$ positive semidefinite

$$\therefore \sigma_i = \sqrt{\lambda_i}, \lambda_i \geq 0$$

° Section 7.2 Problem Sets: 4, 7, 9, 13, 19, 24-25.

4) $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(3x2)(2x3)

$$\det(A^T A - \lambda I) = \begin{vmatrix} -\lambda & 1 & 1 \\ 0 & 2-\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0$$

$$\begin{aligned} &= (0)C_{31} + (1)C_{32} + (-\lambda)C_{33} \\ &= (-1)^5 [(-\lambda)(1)(1)] + (-\lambda)(-1)^6 [(-\lambda)(2-\lambda)(1)] \\ &= (-1)[- \lambda - 1] + (-\lambda)[-2\lambda + \lambda^2 - 1] \\ &= \lambda + 1 + 2\lambda^2 - \lambda^3 + \lambda \\ &= -\lambda^3 + 2\lambda^2 + 2\lambda + 1 \\ &\approx \lambda = -1; \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i; \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

$$A A^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$A^T A$ ° $\det(A A^T - \lambda I) = (2-\lambda)(2-\lambda) - (1)(1) = 0$

$$4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 3 \text{ or } 1.$$

°° $\sigma_1^2 = 3$ $\sigma_2^2 = 1$

$\lambda = 3$ $(A A^T - \lambda I)x = 0$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

set $x_2 = 1$ $-x_1 + (1) = 0$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_1 = 1$$

$\lambda = 1$ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} x = 0$

set $x_2 = -1$ $x_1 + (-1) = 0$

$$x_1 = +1$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

°° $u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$A^T A$: Shared λ s

$\therefore \lambda_1 = 3 \quad \lambda_2 = 1 \quad \lambda_3 = ?$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$\text{trace}(A^T A) = (2) + (2) = 4$

$4 - 3 - 2 = \underline{0}$
 $\lambda_3 = \underline{0}$

~~$\lambda = 3$~~ : ~~$\begin{bmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix} x = 0$~~

~~$\begin{bmatrix} -2 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} x = 0$~~

$\lambda = 3$ $\begin{bmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix} x = 0$

$-2x_1 + x_2 = 0$

set $x_2 = 2$

$\therefore x_1 = 1$

~~$(-2) \cdot x_3 = 0$~~

$x_3 = 1$

$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$\lambda = 1$ $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} x = 0$

$x_2 = 0$

$x_1 + x_3 = 0$

set $x_1 = 1$

$\therefore x_3 = -1$

$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$\lambda = 0$ $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} x = 0$

$x_1 + x_2 = 0$ set $x_2 = -1 \therefore x_1 = 1$

$x_2 + x_3 = 0 \therefore x_3 = 1$

$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} ; v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} ; v_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\therefore AV = U\Sigma$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

(2x3)

(3x3)

(2x2)

(2x3)

memo

$$7) (A^T A) v = \sigma^2 v \quad \text{multiply by } A.$$

$$A(A^T A)v = A\sigma^2 v$$

$$(AA^T)Av = \sigma^2(Av)$$

$$(AA^T)Av = A(A^T A)v = A\lambda v \quad \checkmark$$

If v is an eigenvector of $A^T A$, then $\underline{Av}$ is an eigenvector of AA^T .

$$8) A^T A = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = 0 = \begin{vmatrix} 10-\lambda & 20 \\ 20 & 40-\lambda \end{vmatrix} = (10-\lambda)(40-\lambda) - (20)^2$$

$$= 400 - 50\lambda + \lambda^2 - 400$$

$$= \lambda^2 - 50\lambda$$

$$\lambda = 0 \text{ or } 50.$$

$v_1 (\lambda=50)$:

$$\begin{bmatrix} A^T A - \lambda I \end{bmatrix} x = 0 = \begin{bmatrix} -40 & 20 \\ 20 & -10 \end{bmatrix} x = 0$$

$$-40x_1 + 20x_2 = 0$$

$$\text{set } x_2 = 2; x_1 = 1 \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$v_2 (\lambda=0)$

$$\begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} x = 0$$

set

$$10x_1 + 20x_2 = 0 \quad \text{set } x_1 = 2; x_2 = -1 \quad \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

u_1

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)}{5\sqrt{2} \sqrt{50}}$$

$$= \frac{\begin{bmatrix} 5 \\ 15 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)}{\sqrt{10} \sqrt{50}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 5 \\ 15 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 15 \end{bmatrix} / (5\sqrt{10})$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

u₂

$$u_2 = \frac{AV_2}{\sigma_2} = \frac{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)}{\cancel{\sqrt{10}}}$$

~~$\neq \frac{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}}{\sqrt{10}}$~~

$$= \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)$$

$$u_2 \neq \frac{AV_2}{\sigma_2}$$

$2 > (r=1) \quad 1 \leq r$
for $u_i = \frac{AV_i}{\sigma_i}$

$$u_1 \perp u_2$$

$$u_1^T u_2 = 0$$

$$\left(\frac{1}{\sqrt{10}}\right) \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$\circ \circ \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

$$\circ \circ u_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix}$$

9) $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ 2×2 matrix

rank = 1 $n - r = 2 - 1 = 1 \rightarrow$

Bases:

$$\left. \begin{array}{l} C(A^T) \Rightarrow V_1 \\ \hookrightarrow N(A) \Rightarrow V_2 \\ C(A) \Rightarrow U_1 \\ \hookrightarrow N(A^T) \Rightarrow U_2 \end{array} \right\} \begin{array}{l} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix} \end{array} \quad \checkmark$$

13) $A = U \Sigma V^T$ is square invertible
then $A^{-1} = \underbrace{U}_{\downarrow} \underbrace{\Sigma^{-1}}_{\downarrow} \underbrace{V^T}_{\downarrow}$ ~~V^T~~

$A^{-1} = U \Sigma^{-1} U^T$

~~$A^T A = I$~~

$$\begin{aligned} (U \Sigma^{-1} U^T)(U \Sigma V^T) &= U \Sigma^{-1} (U^T U) \Sigma V^T \\ &= U (\Sigma^{-1} \Sigma) V^T \\ &= U I V^T \\ &= U V^T \quad \checkmark \end{aligned}$$

$\Sigma^{-1} = \text{diag}(\frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_n}) \rightarrow$

σ_i were ordered descending, their reciprocals $\frac{1}{\sigma_i}$ reverse order.

$\sigma_{\max}(A^{-1}) = \frac{1}{\sigma_{\min}(A)}$

$\sigma_{\max}(A^{-1}) \cdot \sigma_{\max}(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} = \kappa(A)$ condition number

$\therefore A$ is a scalar multiple of an orthogonal matrix (always ≥ 1)

$\kappa \rightarrow$ measures how much relative error in b gets amplified when $Ax = b$. Large $\kappa \rightarrow$ answer unreliable regardless of algorithm.

19) A is invertible (with $\sigma_1 > \sigma_2 > 0$)

→ Change A by as small a matrix as possible

↳ to produce a singular matrix A_0 .

$$A = [u_1 \ u_2] \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} [v_1 \ v_2]^T$$

$$A_0 = ?$$

→ Singular means $\sigma_n = 0$

∴ Smallest ?

$$\sigma_1 \gg \sigma_2 \gg \dots \gg \sigma_n$$

∴ Set $\sigma_2 = 0$

for smallest possible
change

$$\therefore A_0 = [u_1 \ u_2] \begin{bmatrix} \sigma_1 & \\ & 0 \end{bmatrix} [v_1 \ v_2]^T$$

Section 8.1 Problem Set: Q6, 8, 11, 12, 16

6) (a) $T(v) = \frac{v}{\|v\|}$

$$T(cv) = \frac{(cv)}{\|cv\|}$$

$$cT(v) = c \frac{v}{\|v\|}$$

$$T(v+w) = \frac{(v+w)}{\|v+w\|}$$

$$\neq T(v) + T(w)$$

$$\frac{v}{\|v\|} + \frac{w}{\|w\|} \neq \frac{v+w}{\|v+w\|}$$

Satisfy: $T(cv) = cT(v)$ ✓

(b) $T(v) = v_1 + v_2 + v_3$

$$T(cv) = c(v_1 + v_2 + v_3)$$

$$= cv_1 + cv_2 + cv_3$$

$$= cT(v)$$

$$= c(v_1 + v_2 + v_3)$$

$$T(v+w) = (v_1 + w_1) + (v_2 + w_2) + (v_3 + w_3)$$

$$= (v_1 + v_2 + v_3) + (w_1 + w_2 + w_3)$$

$$= T(v) + T(w)$$

Satisfy $T(cv) = cT(v)$
 $T(v+w) = T(v) + T(w)$

Linear

(c) $T(v) = (v_1, 2v_2, 3v_3)$

$$T(v+w) = (v_1 + w_1, 2(v_2 + w_2), 3(v_3 + w_3))$$

$$= (v_1, 2v_2, 3v_3) + (w_1, 2w_2, 3w_3)$$

$$= T(v) + T(w)$$

Satisfy $T(cv) = cT(v)$
 $T(v+w) = T(v) + T(w)$

Linear

(d) Satisfy $T(cv) = cT(v)$

$$8) (a) T(v_1, v_2) = (v_1 - v_2, 0)$$

$$\left\{ \begin{array}{l} \text{Range} : \text{line of vectors } (c, 0) \\ \text{Kernel} : \text{line of vectors } (c, c) \end{array} \right\} \checkmark$$

$$(b) T(v_1, v_2, v_3) = (v_1, v_2)$$

$$\begin{array}{l} \text{Range} : \text{plane of vectors } (c, d) \times \mathbb{R}^2 \\ \text{Kernel} : \text{point } (0, 0) \times (0, 0, v_3) \end{array}$$

$$(c) T(v_1, v_2) = (0, 0)$$

$$\begin{array}{l} \text{Range} : (v_1, v_2) \times \{0\} \\ \text{Kernel} : \mathbb{R}^2 \end{array} \checkmark$$

$$(d) T(v_1, v_2) = (v_1, v_1)$$

$$\begin{array}{l} \text{Range} : (v_1, v_1) \times \Delta (1, 1) \\ \text{Kernel} : (0, v_2) \times \begin{array}{c} \text{---} \\ (1, 1) \\ (0, 1) \end{array} \end{array}$$

$$11) T(v) = Av$$

$$A_{m \times n} \quad V_{n \times 1} \quad Av_{m \times 1}$$

$$a) \begin{array}{l} \text{Input: } V \\ \text{Output: } Av \end{array} \quad \begin{array}{l} V \rightarrow R^n \\ W \rightarrow R^m \end{array}$$

$$b) \text{Range: } C(A) \quad \text{Output fills column space.}$$

$$c) \text{Kernal: } N(A) \quad \underline{Av=0}$$

$$12) T: \begin{array}{l} (1,1) \rightarrow (2,2) \\ (2,0) \rightarrow (0,0) \end{array} \quad Av = b$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$(a) v = (2,2) \circ \begin{bmatrix} 4 \\ 4 \end{bmatrix} \checkmark$$

$$(b) v = (3,1) \circ \begin{bmatrix} 2 \\ 2 \end{bmatrix} \checkmark$$

$$(c) v = (-1,1) \circ \begin{bmatrix} 2 \\ 2 \end{bmatrix} \checkmark$$

$$(d) v = (a,b) \circ \begin{bmatrix} b \\ b \end{bmatrix} \times \begin{bmatrix} 2b \\ 2b \end{bmatrix}$$

$$16) T: \mathbb{R}^{M_{2 \times 2}} \rightarrow M_{2 \times 2}^T$$

Find A for $AM = M^T$

$$A \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\begin{bmatrix} x & y \\ z & k \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\begin{aligned} xa + yc &= a \\ xb + yd &= c \end{aligned}$$

$$\begin{aligned} za + kc &= b \\ zb + kd &= d \end{aligned}$$

$$za + k(xb + yd) = b$$

$$za + kxb + kyd = b$$

$$za + kyd = (1 - kx)b$$

No A gives $A \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

no solution

→ Dimension of space not shape of elements determines size of matrix. for A

→ $T(v) = Av \rightarrow$ is one representation and not the def of linear.

→ Definition is preserving combinations.

→ Whether it is expressible ~~at~~ on the left as AM depends on how the inputs are packaged.